Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
  - Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

● **Data Loading**: Imported the dataset using `pandas.`

● **Initial Exploration:** Used `df.info()` to check structure and `.describe(include = "all")` for summary statistics.

```
[31]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 19 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customer_id             3900 non-null   int64
 1   age                     3900 non-null   int64
 2   gender                  3900 non-null   object
 3   item_purchased          3900 non-null   object
 4   category                3900 non-null   object
 5   purchase_amount         3900 non-null   int64
 6   location                3900 non-null   object
 7   size                    3900 non-null   object
 8   color                   3900 non-null   object
 9   season                  3900 non-null   object
 10  review_rating           3900 non-null   float64
 11  subscription_status     3900 non-null   object
 12  shipping_type           3900 non-null   object
 13  discount_applied        3900 non-null   object
 14  previous_purchases      3900 non-null   int64
 15  payment_method          3900 non-null   object
 16  frequency_of_purchases  3900 non-null   object
 17  age_group               3900 non-null   category
 18  purchase_frequency_days 3900 non-null   int64
dtypes: category(1), float64(1), int64(5), object(12)
memory usage: 552.6+ KB
```

```
[30]: df.describe(include="all")
```

[30]:

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping_ty
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3900.000000	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shippi
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	6
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750051	NaN	Na
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.713590	NaN	Na
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	Na
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	Na
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	Na
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	Na
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	Na

- **Missing Data Handling:**
Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.

- **Column Standardization:** Renamed columns to snake case for better readability and documentation.

- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:**
Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used.`

- **Database Integration:**
Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions):

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared the total revenue generated by male vs. female customers.

Result Grid | Filter Rows:

gender	reveneu
Male	157890
Female	75191

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

Result Grid | Filter Rows: | Export:

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94

customer_data 6

3. Top 5 Products by Rating – Found products with the highest average review ratings.

Result Grid | Filter Rows: | Export:

item_purchased	avg_rating
Gloves	3.8614285714285725
Sandals	3.8443750000000003
Boots	3.8187500000000005
Hat	3.8012987012987005
Skirt	3.784810126582278

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

Result Grid | Filter Rows:

shipping_type	avg_purchase_type
Express	60.4752
Standard	58.4602

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

Result Grid | Filter Rows: | Export: | Wr

subscription_status	total_customer	avg_spend	total_reveneu
Yes	1053	59.4919	62645
No	2847	59.8651	170436

6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

item_purchased	total_purchased	discount_purchases	discount_percentage
Hat	154	77	50.0000
Sneakers	145	72	49.6552
Coat	161	79	49.0683
Sweater	164	79	48.1707
Pants	171	81	47.3684

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

Result Grid | Filter Rows:

customer_segment	total_customers
Loyal	3116
Returning	701
New	83

8. Top 3 Products per Category – Listed the most purchased products within each category.

Result Grid | Filter Rows: | Export: | Wrap Ce

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Checked whether customer with >5 purchases are more likely to subscribe.

Result Grid | Filter Rows:

subscription_status	repeat_buyers
Yes	958
No	2518

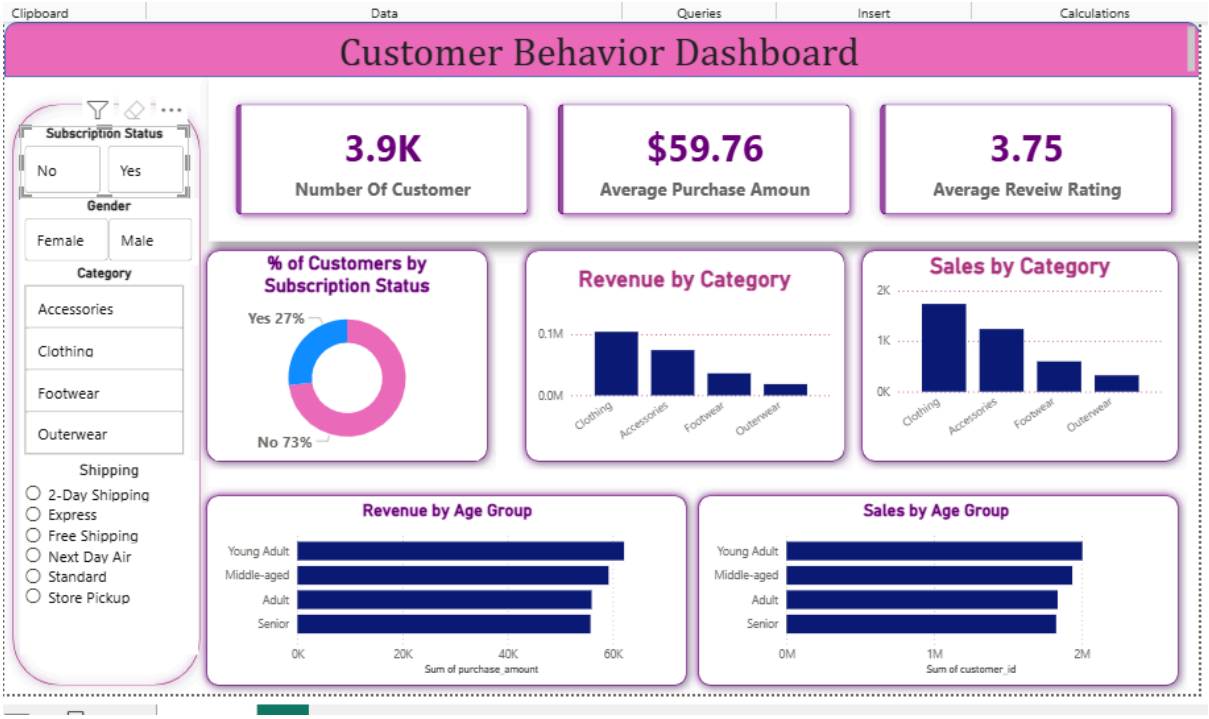
10. Revenue by Age Group – Calculated total revenue contribution of each age group.

Result Grid | Filter Rows:

age_group	total_reveneu
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

● **Boost Subscriptions** – Promote exclusive benefits for subscribers.
● **Customer Loyalty Programs** – Reward repeat buyers to move them into the "Loyal" segment.
● **Review Discount Policy** – Balance sales boosts with margin control.
● **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
● **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.